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EP001: Physical System Description

Double-Inverted Pendulum Control Theory

Episode ID: EP001 Difficulty: Intermediate Duration: 2 h

Double-Inverted Pendulum Learning Module

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Contents

section0 Introduction

subsection0.0 Overview

The **double-inverted pendulum (DIP)** is one of the canonical benchmarks in nonlinear control theory. A single horizontal force drives a cart along a track; two rigid links are hinged in series at the cart and must be balanced upright using that single actuator.

- The upright equilibrium is open-loop unstable.
- The system is *underactuated*: two rotational degrees of freedom, one control input.
- Strong nonlinear coupling exists between the two pendulum links.
- The system is a standard testbed for sliding mode control (SMC), model predictive control (MPC), and reinforcement learning.

subsection0.0 System Characteristics

definitionA mechanical system is underactuated *if the number of independent control inputs is strictly less than the number of degrees of freedom.*

recordtarget The DIP satisfies this definition with three degrees of freedom (cart position, link-1 angle, link-2 angle) and one control input (cart force). Key dynamical properties are listed below.

enumi**Nonlinear coupling**: the inertia matrix depends on joint angles, creating state-dependent effective mass.

0. enumi**Unstable equilibrium**: small perturbations cause the pendulums to fall without active feedback.
0. enumi**Non-minimum phase tendencies**: zero dynamics near upright may exhibit unstable behavior depending on output choice.
0. enumi**Disturbance sensitivity**: external impulses propagate rapidly through the kinematic chain.

section0 Physical System Description

subsection0.0 System Configuration

The DIP hardware consists of the following components.

- **Cart** (mass m_0): moves along a horizontal track; travel limited to $\pm 1\,\mathrm{m}$ in simulation.
- **Link 1** (mass m_1, length l_1): rigid rod, hinged at the cart top; angle θ_1 measured from the upright vertical.
- **Link 2** (mass m_2, length l_2): rigid rod, hinged at the tip of Link 1; angle θ_2 measured from the upright vertical.
- **Actuator**: motor applies horizontal force u to the cart.
- **Sensing**: optical encoders measure x, θ_1, θ_2; velocities are estimated by finite differences.

remarkAngles are measured from the upright position (counterclockwise positive), consistent with the Euler-Lagrange derivation in Section .

```
recordtarget      [m2, l2]   Link 2
          |  theta_2
     [m1, l1]   Link 1
          |  theta_1
  ========================  Track
       [ m0 ]   Cart
        <--- u (control force)
```

subsection0.0 Physical Constraints

Operating bounds that any controller must respect:

enumiCart position: $x \in [-1, 1]\,\mathrm{m}$

0. enumiControl force magnitude: $|u| \le u_{\max} = 50\,\mathrm{N}$
0. enumiLink angles: unconstrained in simulation; hardware stops near $\theta_i = \pm\pi$

The mass ordering $m_0 > m_1 > m_2$ ensures the heavier cart provides the necessary reaction force for upright balance. The length ordering $l_1 > l_2$ gives the base link greater control authority.

section0 Mathematical Model

subsection0.0 State Vector Definition

definitionThe system state vector is

$$equation\mathbf{x} = \left[x,\ \theta_1,\ \theta_2,\ \dot{x},\ \dot{\theta}_1,\ \dot{\theta}_2\right]^T \in \mathbb{R}^6, \tag{0}$$

where x is the cart position (m), θ_i are link angles (rad) from the upright, and dots denote time derivatives.

recordtarget The generalized coordinate vector is $\mathbf{q} = [x,\ \theta_1,\ \theta_2]^T$.

subsection0.0 Equations of Motion

The equations of motion are derived via the **Euler-Lagrange method**. Let $L = T - V$ denote the Lagrangian. Applying

$$equation\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) - \frac{\partial L}{\partial q_i} = Q_i, \quad i = 1, 2, 3, \tag{0}$$

where the generalized forces are $Q_1 = u$, $Q_2 = Q_3 = 0$, yields the standard manipulator form

$$equation\mathbf{M}(\mathbf{q})\,\ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\,\dot{\mathbf{q}} + \mathbf{G}(\mathbf{q}) + \mathbf{F}\,\dot{\mathbf{q}} = \mathbf{B}\,u + \mathbf{d}(t), \tag{0}$$

where $\mathbf{F} = \mathrm{diag}(b_0, b_1, b_2)$ is the viscous friction matrix, $\mathbf{B} = [1, 0, 0]^T$, and $\mathbf{d}(t)$ represents bounded external disturbances.

subsection0.0 Inertia Matrix

The inertia matrix $\mathbf{M}(\mathbf{q}) \in \mathbb{R}^{3\times3}$ is symmetric and configuration-dependent:

$$equation\mathbf{M}(\mathbf{q}) = \begin{bmatrix} M_{11} & M_{12} & M_{13} \\ M_{12} & M_{22} & M_{23} \\ M_{13} & M_{23} & M_{33} \end{bmatrix}, \tag{0}$$

with elements derived from the kinetic energy:

$$M_{11} = m_0 + m_1 + m_2, \quad \text{equation}(0)$$
$$M_{12} = (m_1 r_1 + m_2 l_1)\cos\theta_1 + m_2 r_2 \cos\theta_2, \quad \text{equation}(0)$$
$$M_{13} = m_2 r_2 \cos\theta_2, \quad \text{equation}(0)$$
$$M_{22} = m_1 r_1^2 + m_2 l_1^2 + I_1, \quad \text{equation}(0)$$
$$M_{23} = m_2 l_1 r_2 \cos(\theta_1 - \theta_2) + I_2, \quad \text{equation}(0)$$
$$M_{33} = m_2 r_2^2 + I_2, \quad \text{equation}(0)$$

where r_i is the distance from joint i to the center of mass of link i.

theoremThe inertia matrix $\mathbf{M}(\mathbf{q})$ *is positive definite for all physically admissible configurations, i.e.,* $\mathbf{M}(\mathbf{q}) \succ 0$.

recordtarget subsection**0.0 Coriolis and Centrifugal Matrix**

$\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})$ is computed from Christoffel symbols:

$$\textit{equation} C_{ij} = \sum_{k=1}^{3} \Gamma_{ijk}\, \dot{q}_k, \qquad \Gamma_{ijk} = \frac{1}{2}\left(\frac{\partial M_{ij}}{\partial q_k} + \frac{\partial M_{ik}}{\partial q_j} - \frac{\partial M_{jk}}{\partial q_i} \right). \quad (0)$$

This construction ensures $\dot{\mathbf{M}} - 2\mathbf{C}$ is skew-symmetric, a property used in passivity-based stability proofs.

subsection**0.0 Gravity and Friction Terms**

The gravity vector follows from potential energy $V = m_1 g r_1 \cos\theta_1 + m_2 g (l_1 \cos\theta_1 + r_2 \cos\theta_2)$:

$$\textit{equation} G_1 = 0, \qquad G_2 = -(m_1 r_1 + m_2 l_1) g \sin\theta_1, \qquad G_3 = -m_2 r_2 g \sin\theta_2. \quad (0)$$

section0 Nonlinearity Analysis

subsection0.0 Configuration-Dependent Inertia

The off-diagonal elements M_{12}, M_{13}, M_{23} depend on trigonometric functions of the joint angles:

0. enumiM_{12} varies by up to 40% as θ_1 sweeps from 0 to $\pi/4$ rad (nominal parameters).
0. enumiM_{23} varies by up to 35% as $(\theta_1 - \theta_2)$ changes, representing coupling between the two links.
0. enumiNear collinear configurations the condition number of $\mathbf{M}(\mathbf{q})$ increases, potentially causing ill-conditioning in the numerical inversion required to compute $\ddot{\mathbf{q}}$.

Gains tuned at the upright equilibrium $\mathbf{q} = \mathbf{0}$ may therefore be suboptimal at large angles, motivating gain scheduling or robust designs.

subsection0.0 Trigonometric Nonlinearity

theoremThe gravity vector $\mathbf{G}(\mathbf{q})$ *is globally bounded:* $\|\mathbf{G}(\mathbf{q})\| \le G_{\max}$ *for all* $\mathbf{q}$*, where* $G_{\max}$ *depends only on system masses, link lengths, and* g.

recordtarget Dominant nonlinear terms are $\sin\theta_i$ in $\mathbf{G}$ and $\cos(\theta_1 - \theta_2)$ in M_{23}. For small angles ($|\theta_i| \le 0.3$ rad) the approximations $\sin\theta_i \approx \theta_i$ and $\cos\theta_i \approx 1$ reduce the system to a linear time-invariant plant suitable for LQR or pole-placement design.

section0 System Parameters

subsection0.0 Nominal Parameter Values

Table lists the nominal parameters used throughout this learning module and in the accompanying simulation code.

table

Table 0: Nominal DIP System Parameters

Parameter	Symbol	Value	Unit
Cart mass	m_0	1.50	kg
Link 1 mass	m_1	0.20	kg
Link 2 mass	m_2	0.15	kg
Link 1 length	l_1	0.40	m
Link 2 length	l_2	0.30	m
Link 1 COM distance	r_1	0.20	m
Link 2 COM distance	r_2	0.15	m
Link 1 moment of inertia	I_1	0.0081	$\mathrm{kg\,m^2}$
Link 2 moment of inertia	I_2	0.0034	$\mathrm{kg\,m^2}$
Gravitational acceleration	g	9.81	$\mathrm{m\,s^{-2}}$
Maximum control force	$u_{\max}$	50	N
Sampling period	Δt	0.01	s

The inertia values satisfy the parallel-axis theorem constraint $I_i \geq m_i r_i^2$ (equality for uniform rods).

subsection0.0 Parameter Uncertainty

Real hardware deviates from nominal values due to:

0• Manufacturing tolerances in link lengths ($\pm 2\,\mathrm{mm}$ typical).
- Mass variation with attached payload or wear.
- Friction coefficients that change with temperature and lubrication state.
- Encoder resolution limiting velocity-estimation accuracy.

Robust SMC designs must guarantee stability across the uncertainty polytope $\mathcal{P} = \{p \mid p_{\min} \leq p \leq p_{\max}\}$ defined by worst-case parameter deviations.

section0 Linearized Model

subsection0.0 Equilibrium Points

definitionA state $\mathbf{x}_e$ *is an* equilibrium *of () if* $\dot{\mathbf{q}}_e = \mathbf{0}$*,* $\ddot{\mathbf{q}}_e = \mathbf{0}$*, and* $u_e = 0$.

recordtarget Two equilibria are of practical interest:

enumi**Upright (unstable)**: $\mathbf{x}_e = \mathbf{0}$ – the control objective of this module.

0. enumi**Downward (stable)**: $\theta_1 = \theta_2 = \pi$ – natural rest position; swing-up controllers target this as a starting condition before handing off to the balancing controller.

subsection0.0 Linearized State-Space Representation

Linearizing () about $\mathbf{x}_e = \mathbf{0}$ via a first-order Taylor expansion gives

$$equation\dot{\mathbf{x}} = \mathbf{A}\,\mathbf{x} + \mathbf{b}\,u, \tag{0}$$

where $\mathbf{A} \in \mathbb{R}^{6\times 6}$ and $\mathbf{b} \in \mathbb{R}^6$ are constant matrices evaluated at $\mathbf{q} = \mathbf{0}$.

Numerical analysis with the nominal parameters of Table gives:

0• Two open-loop eigenvalues with positive real parts (unstable pendulum modes).

- Four eigenvalues with negative real parts (stable cart and damped oscillatory modes).
- Controllability matrix rank equal to 6, confirming full controllability.

section0 Numerical Properties

subsection0.0 State Scaling

Raw state units span several orders of magnitude: position in metres, angles in radians, velocities up to $\sim 10\,\mathrm{rad/s}$ during aggressive manoeuvres. Normalized coordinates

$$equation\tilde{x} = \frac{x}{x_{\max}}, \quad \tilde{\theta}_i = \frac{\theta_i}{\theta_{\max}}, \quad \tilde{\dot{x}} = \frac{\dot{x}}{\dot{x}_{\max}}, \tag{0}$$

improve the conditioning of PSO cost functions and observer design.

subsection0.0 Time Scaling

enumiThe highest linearized natural frequency sets the Nyquist requirement; $\Delta t = 0.01\,\mathrm{s}$ (100 Hz) provides adequate margin for nominal parameters.

0. enumiFixed-step Runge-Kutta 4 integration with this timestep yields energy-conservation error below 0.1% over 10 s with zero friction and zero control input.
0. enumiVariable-step integrators should use absolute and relative tolerances of 10^{-6} to resolve fast pendulum modes correctly.

section0 Summary and Key Takeaways

subsection0.0 System Overview

The DIP is a 6-state, 1-input nonlinear plant. Its equations of motion have the standard manipulator form $\mathbf{M}\ddot{\mathbf{q}} + \mathbf{C}\dot{\mathbf{q}} + \mathbf{G} = \mathbf{B}u$, which is directly exploited by energy-based and sliding-mode controllers. The system is controllable, unstable in open loop, and well-suited as a benchmark for advanced control algorithms.

subsection0.0 Key Challenges

0. enumi**Nonlinearity**: inertia and gravity terms are angle-dependent; linear designs lose validity beyond $\pm 0.3\,\mathrm{rad}$.
0. enumi**Coupling**: cart motion excites both links; second-link motion back-drives the first through M_{23}.
0. enumi**Underactuation**: one force must simultaneously regulate three coordinates, requiring exploitation of internal dynamics.
0. enumi**Parameter uncertainty**: friction and mass vary in hardware; robust designs must cover the full uncertainty polytope.

subsection**0.0 Control Implications**

theoremFor system () with $\mathbf{M}(\mathbf{q}) \succ 0$ *and bounded disturbance* $\|\mathbf{d}(t)\| \leq D$*, a sliding-mode controller* $u = -\eta \operatorname{sgn}(\sigma)$ *with gain* $\eta > D/\lambda_{\min}(\mathbf{M})$ *drives the sliding variable* σ *to zero in finite time.*

recordtarget• Linear controllers (LQR, PID) are effective near the upright equilibrium but require gain scheduling for large-angle operation.

- Sliding-mode and adaptive controllers provide robustness to parameter uncertainty and bounded disturbances.
- PSO-tuned gains consistently improve ISE, IAE, and chattering index over manually tuned baselines.
- Hardware implementations require constraint-aware tuning to respect force limits and cart travel bounds.